

Yunsung Chung

Computer Science Ph.D. Researcher | Applied Machine Learning

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EXPERIENCE

Samsung Research America – Research Intern

May 2025 – Aug 2025

Mountain View, CA / Supervisor: Dr. Sharanya Desai

- Designed a **biomarker-specific temporal-decay loss** to align dense smartwatch pulse-wave (PPG) signals with infrequent clinical labels, accounting for biomarker-specific label staleness.
- Built a PPG signal-quality index (SQI) algorithm applied to **more than 80,000 hours of PPG from 50,000 participants**, enabling task-specific filtering and consistent data-quality checks across experiments.
- Owned temporal windowing, model training, and subject-wise evaluation; worked with research scientists and engineers to align the evaluation protocol with wearable sensing constraints.

Tulane University, TRIAD Lab – Research Assistant

Aug 2022 – Present

New Orleans, LA / Supervisor: Dr. Jihun Hamm

- Developed **SOFA**, a deep-learning framework for simulating and optimizing atrial fibrillation ablation strategies (MICCAI 2025).
- Developed an intervention-aware world model of longitudinal patient-state evolution after ablation (MWM 2026, oral), with cardiology collaborators.
- Built and released **EHRDyn**, a five-task benchmark for retrospective healthcare reinforcement learning, with reusable submission and scoring interfaces covering 21 method specifications and six off-policy estimators. Analyzed estimator support, uncertainty, value error, and policy-ranking stability across complementary simulators.
- Developed **CRAFT**, reward-aligned diffusion adaptation across four medical-imaging domains; earned **55.3% of top-choice votes vs. 26.3% for TI+LoRA** in a four-method blinded comparison (400 cases; two physician coauthors). Combined structured clinical checklists and multimodal evaluators with diffusion finetuning; tested downstream utility, evaluator dependence, and memorization.
- Built **EHRGym**, a synthetic clinical benchmark for offline reinforcement learning, to diagnose failures caused by partial observation, irregular measurement timing, and repetitive logging.
- Contributed to the development of ECG-to-PPG representation transfer through contrastive learning and teacher-student distillation using temporally paired recordings, including preprocessing, denoising, quality screening, internal/external cohort evaluation for PPG-only rhythm inference, and integration into app-based research prototypes.
- Led a **130+ study** review and empirical benchmark of synthetic-image utility and privacy, including membership-inference risk; published at USENIX Security 2025.

Airport Railroad Express – Hardware Engineer

Nov 2017 – May 2019

Incheon, South Korea

- Tested and debugged automated fare-collection and passenger-service hardware for a high-reliability transportation system.

EDUCATION

Tulane University

Ph.D. in Computer Science (in progress)

Aug 2022 – Present

New Orleans, LA

The Australian National University

M.S. in Computer Science

Jul 2019 – Jul 2021

Canberra, Australia

Dongguk University

B.S. in Electronics, Information & Communication Engineering

Mar 2012 – Feb 2018

Seoul, South Korea

TECHNICAL SKILLS

Tools: Python, PyTorch, Hugging Face, NumPy, SciPy, pandas, Scikit-learn, SQL, Git, AWS, Weights & Biases.

Methods: Time-series models, Transformers, multimodal learning, diffusion models, LoRA/PEFT, DPO-based diffusion adaptation, contrastive learning, offline RL, off-policy evaluation, uncertainty estimation.

Data: Longitudinal EHR, ECG/PPG, medical imaging, sparse clinical labels.

RESEARCH SOFTWARE

- **ResearchWiki / ResearchOMX:** Built an LLM-assisted research workflow with 41 automated research-consistency checks, links from claims to supporting evidence, and five separate review roles. Used Git and a Markdown wiki to preserve experiment records and auditable research decisions. Human approval controls experiment execution and changes to research conclusions.
- **EHRDyn public benchmark:** Packaged task definitions, scoring interfaces, validation tests, and reproducible reports so research methods can be compared under the same evaluation protocol. Implemented versioned task contracts, synthetic smoke tests, and explicit unsupported-result states; kept results tied to fixed data roles and documented reproducible comparison workflows.

SELECTED PUBLICATIONS

Peer-Reviewed Publications

- [1] Yingshuo Liu, Chanhho Lim, Ghassan Bidaoui, Christian Massad, **Yunsung Chung**, et al. *Reclassifying Single Lead Sinus Rhythm ECGs for Improved AF Recurrence Detection With Deep Learning*. **Circulation: Arrhythmia and Electrophysiology**, e014659, 2026. doi:10.1161/CIRCEP.125.014659.
- [2] **Yunsung Chung**, Yingshuo Liu, Abboud F. Hassan, Han Feng, Mary M. Maleckar, Nassir Marrouche, and Jihun Hamm. *Intervention-Aware Clinical World Model for Post-Op Outcome Forecasting in Cardiology*. **MICCAI Workshop on Medical World Models (MWM)**, 2026. **Oral presentation**.
- [3] **Yunsung Chung**, Keum San Chun, Migyeong Gwak, Han Feng, Yingshuo Liu, Chanhho Lim, Viswam Nathan, Nassir Marrouche, and Sharanya Arcot Desai. *Weighted Temporal Decay Loss for Learning Wearable PPG Data with Sparse Clinical Labels*. **ICASSP**, 2026.
- [4] **Yunsung Chung**, Chanhho Lim, Ghassan Bidaoui, Christian Massad, Nassir Marrouche, and Jihun Hamm. *SOFA: Deep Learning Framework for Simulating and Optimizing Atrial Fibrillation Ablation*. **MICCAI**, 2025.
- [5] **Yunsung Chung**, Yunbei Zhang, Nassir Marrouche, and Jihun Hamm. *SoK: Can Synthetic Images Replace Real Data? A Survey of Utility and Privacy of Synthetic Image Generation*. **USENIX Security**, 2025.
- [6] **Yunsung Chung**, Janet Wang, and Jihun Hamm. *Bridging the Gap: Synthetic Data Augmentation through Inversion and Distribution Matching for Few-Shot Learning*. **AI-ML Systems**, 2024.
- [7] Janet Wang, **Yunsung Chung**, Zhengming Ding, and Jihun Hamm. *From Majority to Minority: A Diffusion-Based Augmentation for Underrepresented Groups in Skin Lesion Analysis*. **MICCAI Workshop**, 2024.
- [8] **Yunsung Chung**, Chanhho Lim, Chao Huang, Nassir Marrouche, and Jihun Hamm. *FBA-Net: Foreground and Background Aware Contrastive Learning for Semi-Supervised Atrium Segmentation*. **MICCAI Workshop**, 2023.

Manuscripts Under Review

- [9] **Yunsung Chung**, Alex El Darzi, Carlo El Khoury, Han Feng, Nassir Marrouche, and Jihun Hamm. *CRAFT: Three-Signal Reward Adaptation for Medical Image Synthesis*. Under review at **ICLR**, 2027.
- [10] **Yunsung Chung**, Han Feng, Nassir Marrouche, and Jihun Hamm. *EHRGym: A Diagnostic Benchmark for Offline Reinforcement Learning*. Under review at **ICLR**, 2027.
- [11] **Yunsung Chung**, Han Feng, Nassir Marrouche, and Jihun Hamm. *EHRDyn: A Benchmark for Offline Reinforcement Learning from EHRs with Complementary Simulators*. Under review at **KDD**, 2027.
- [12] Yingshuo Liu, Alex El Darzi, **Yunsung Chung**, et al. *Deep Learning-Based Translation from Wearable ECG to PPG for PPG-Only Cardiac Rhythm Classification*. Under review at **npj Digital Medicine**, 2026.

AWARDS AND SERVICE

Kaggle: Competitions Expert with four bronze medals across 10 completed competitions (seven solo, three team): Image Matching 2022 (96/642), Sartorius (116/1,505), Happywhale (127/1,588), and PetFinder Pawpularity (238/3,537).

Reviewer: NeurIPS 2026; ICML 2026; MICCAI 2024–2026; IEEE TCSVT, 2023.

Robotics: FIRA RoboWorld Cup 2016, 1st Place; International Robot Contest 2016, 2nd Place.

Outreach and Scholarships: NeurIPS 2023 College Outreach Volunteer; First-Class Academic Excellence Scholarship, Dongguk University, 2012 and 2015–2017.